

WTVB01-485 Datasheet and User Guide



Product Specification Sheet:

Model: WT-VB01-485

Description: Vibration sensor

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1. Product Overview

1. The module has a built-in voltage stabilization circuit, with an operating voltage of 9V~36V, and is easy to connect.
2. Advanced digital filtering technology is adopted, which can effectively reduce measurement noise and improve measurement accuracy.
3. We also provide users with all the necessary host computers, user manuals, and development manuals.
4. Supports RS-485 interface. Allows users to choose the optimal connection method. Serial port speed is adjustable from 4800bps to 230400bps.
5. Application areas: It can be widely used for bearing vibration measurement and real-time monitoring of rotating machinery such as submersible pumps, fans, steam turbines, coal mills, oxygen generators, generators, centrifuges, compressors, water pumps, and motors.
6. Three-axis displacement, three-axis velocity, and three-axis frequency outputs meet users' needs for comprehensive vibration and impact measurement, determining whether the measured object (motor, water pump) is damaged. If machine malfunctions are caused by factors such as bearing wear, bearing cracking, poor dynamic balance, or misalignment, the vibration sensor can detect the fault in advance and issue a warning, preventing the machine from continuing to operate under adverse conditions and causing further damage and economic losses.
7. Multiple installation methods: magnetic connection and threaded connection. Secure and stable, easy and convenient to install and disassemble.
8. Double-ended stud bolt with both forward and reverse threads: Stainless steel external hexagonal double-ended stud bolt with both



forward and reverse threads, double-ended stud, left-hand and right-hand dual-thread screw.

2. Parameter Indicators

3. Basic parameters

parameter	condition	Minimum value	default	Maximum value
Communication interface	485 interface	4800bps	9600bps	230400bps
Output content	On-chip time, 3-axis acceleration, 3-axis vibration velocity, 3-axis vibration displacement, 3-axis vibration frequency, temperature			
range	Vibration velocity: 0~600mm/s, vibration displacement: 0~30000um, vibration frequency: 5-100Hz			
accuracy	<FS±4%			
weight			158.17g	
Operating temperature		-40℃		85℃
Storage temperature		-40℃		85℃
Impact resistance				20000g
Protection level	IP68			

3.1. Parameter Description

1. Output content

- Vibration velocity: the speed at which the point of vibration moves at the moment of vibration of an object.
- Vibration displacement: amplitude, the amplitude of the shaking at the point of vibration at the moment of vibration of an object.
- Vibration frequency: The number of vibrations an object makes per unit time.
- Temperature: Chip operating temperature, not external temperature

2. Accuracy

<FS±4%: This indicates 4% of the measurement range. Taking vibration displacement as an example, when the actual displacement value is 300um, the data error is 4% of 300um. When the actual value is 3000um, the data error is 4% of 3000um.

Note: The product has requirements for vibration frequency and is not suitable for low-frequency vibration scenarios. A vibration frequency of at least 5Hz is required to ensure data reliability.

3. Detection cycle (use default parameters, cannot be changed at this time)

The parameters for chip data processing, such as setting it to 100Hz, mean that the chip processes 100 data entries per second.

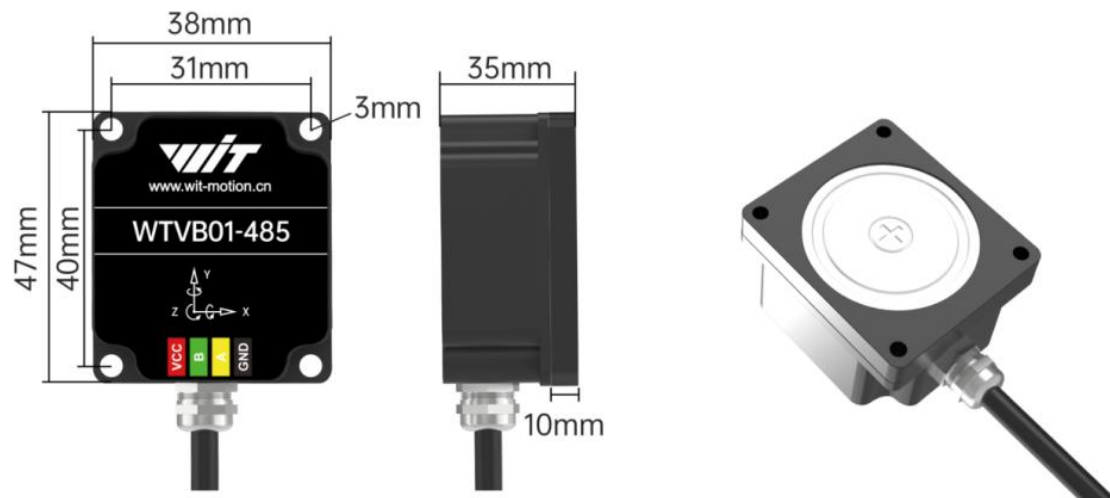
4. Cutoff frequency (use default parameters, cannot be changed at this time)

The cutoff frequency is the frequency at which the sensor can recognize vibration frequencies above a certain threshold. For example, if you set the cutoff frequency to 10Hz, the sensor will filter out vibration frequencies below 10Hz.

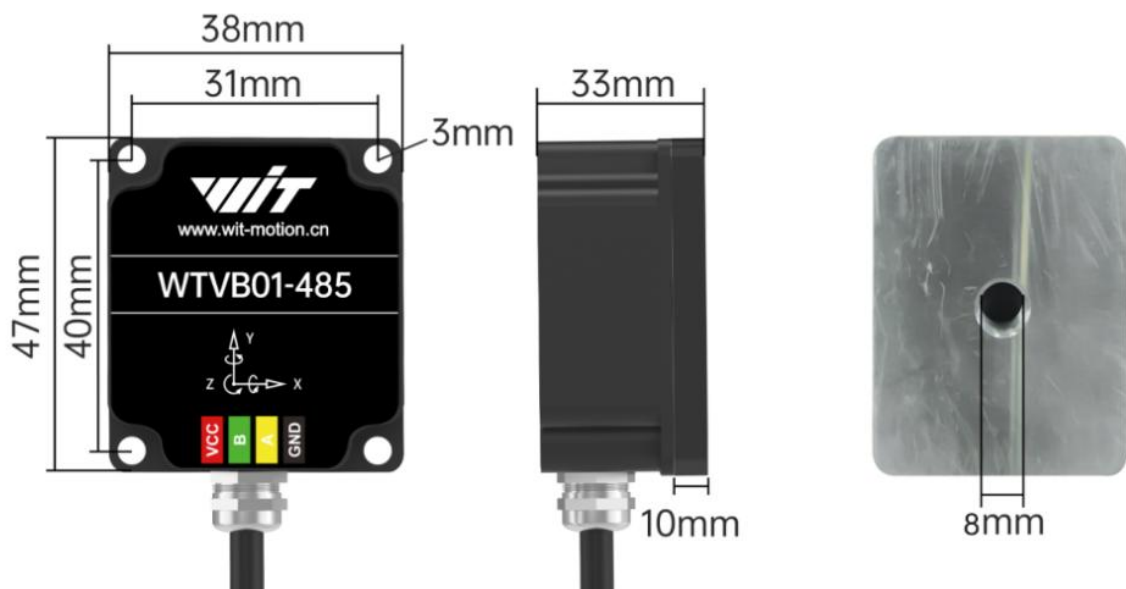
If the cutoff frequency is set to 10Hz and the detection period is set to 100Hz, then the output will be the data with a vibration frequency higher than 10Hz out of 100 data points per second.

3.2. Product Dimensions

Magnetic installation dimensions:



Threaded mounting dimensions:



3.3. Parameter comparison

Product Appearance					
Model		WTVB02-485	WTVB01-485	WTVB05	WTVB01-BT50
Output	Vibration acceleration	✓	✓	✓	/
	Vibration speed	✓	✓	✓	✓
	Vibration displacement	✓	✓	✓	✓
	Vibration frequency	✓	✓	✓	✓
	Chip Time	✓	✓	✓	/
	Chip temperature	✓	✓	Ambient temperature	✓
Output rate	Normal mode (1-200Hz)	Send a command and return once(1-200Hz)			1-100Hz (default 20Hz)
Range	Vibration acceleration	±16G	±16G	±16G	/
	Vibration displacement	0~60000um	0-30000um	0-60000um	0-30000um
	Vibration speed	0~600mm/s	0-600mm/s	0-600mm/s	0-600mm/s
	Vibration frequency	5-2000Hz	5-1000Hz	5-2000Hz	5-100Hz
Accuracy		<F.S±4%	<F.S±4%	<F.S±4%	<F.S±5%
Chip sampling rate		16KHz	1KHz	16KHz	1KHz
Communication interface		RS485	RS485	RS485/CAN	BLE5.0/TYPE-C
Communication Protocol		Modbus protocol	Modbus protocol	Modbus protocol CAN protocol	BLE5.0 protocol
Baud rate	RS485	4800bps-230400bps (default 9600bps)			115200bps (fixed)
	CAN	/	/	3K-1M(default 250K)	/
Temperature	Operating temperature	-40°C~85°C			-20°C~+60°C
	Storage temperature	-40°C~85°C			
Waterproof grade		IP67			Not waterproof
Operating voltage		9-36V			5V/3.7V(Battery voltage)
Working current		8mA		12mA (12V)	5mA(3.7V)
Weight		182g		67.5g	20g
Size		47*38*35mm			51.5*36.1*15mm

3.4. Electrical parameters

parameter	condition	Min	Default	Max
Power supply voltage		9V	12V	36V
Operating current	Working (12V)		8mA	
Power consumption			96mW	

4. Hardware connection method

5. Wiring method

Wiring	RED	YELLOW	GREEN	BLACK
	VCC9-36V	A	B	GND
	Positive pole of the power supply			Negative pole of the power supply



5.1. Installation method

Magnetic attachment or threaded mount
Super stable, easy to install and remove



Magnetic Attachment

Magnet attaches to the device to detect vibration
Super suction up to 70kg.

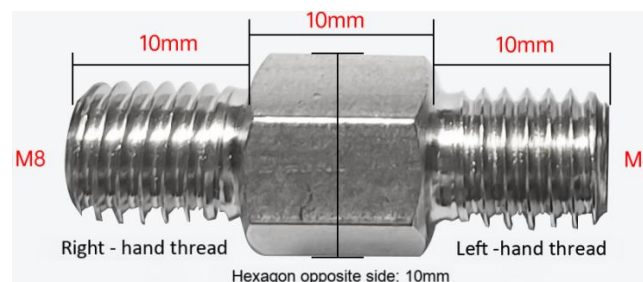




Threaded Mount

Default thread size is M8*1.25mm.
Stable threaded mounting





6. Software usage instructions

7. PC software download

1. CH340 driver download link:

https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

2. Witmotion software download link:

https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

7.1. Connect to the PC software

to the computer via a USB adapter , open Witmotion.exe , and after installing the CH340 driver, you can query the corresponding port number in the device manager, as shown in the figure:

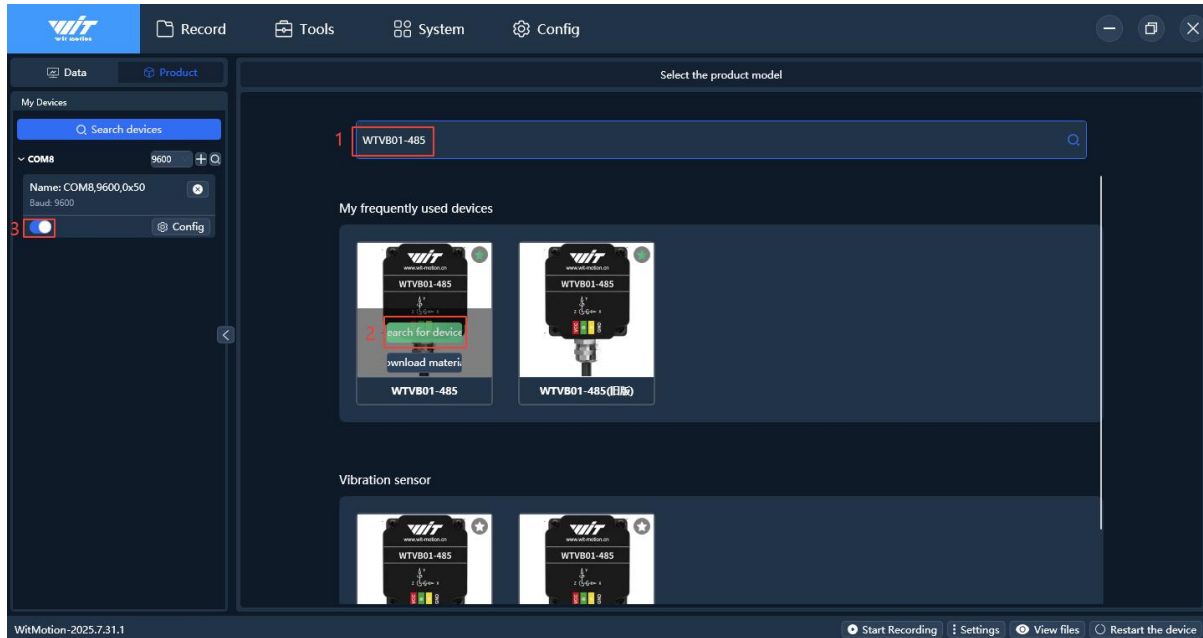


After the device is connected, open the Witmotion.exe software.

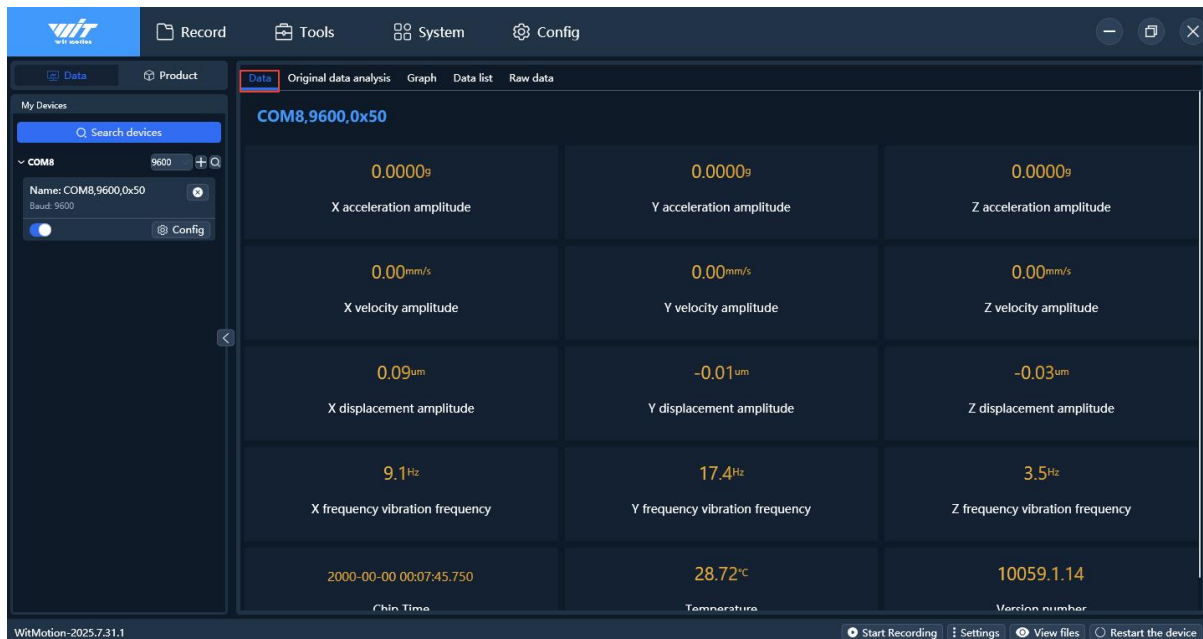
名称	修改日期	类型	大小
AutoUpdateApp	2024/9/6 星期五 17:21	文件夹	
Bin	2025/4/8 星期二 12:05	文件夹	
Config	2025/4/29 星期二 14:...	文件夹	
Plugins	2024/9/6 星期五 17:21	文件夹	
Record	2025/4/29 星期二 14:...	文件夹	
Temp	2024/9/6 星期五 17:24	文件夹	
1.txt	2025/4/18 星期五 19:...	文本文档	1 KB
debug.log	2025/4/18 星期五 19:...	文本文档	1 KB
WitMotion.exe	2025/4/16 星期三 17:...	应用程序	1,543 KB
WitMotion.exe.config	2025/4/16 星期三 17:...	CONFIG 文件	19 KB
WitMotion.pdb	2025/4/16 星期三 17:...	PDB 文件	424 KB

7.1.1. Automatic Search

1. Select the model number WTVB01-485.
2. Click Search device.
3. Check the serial port number.



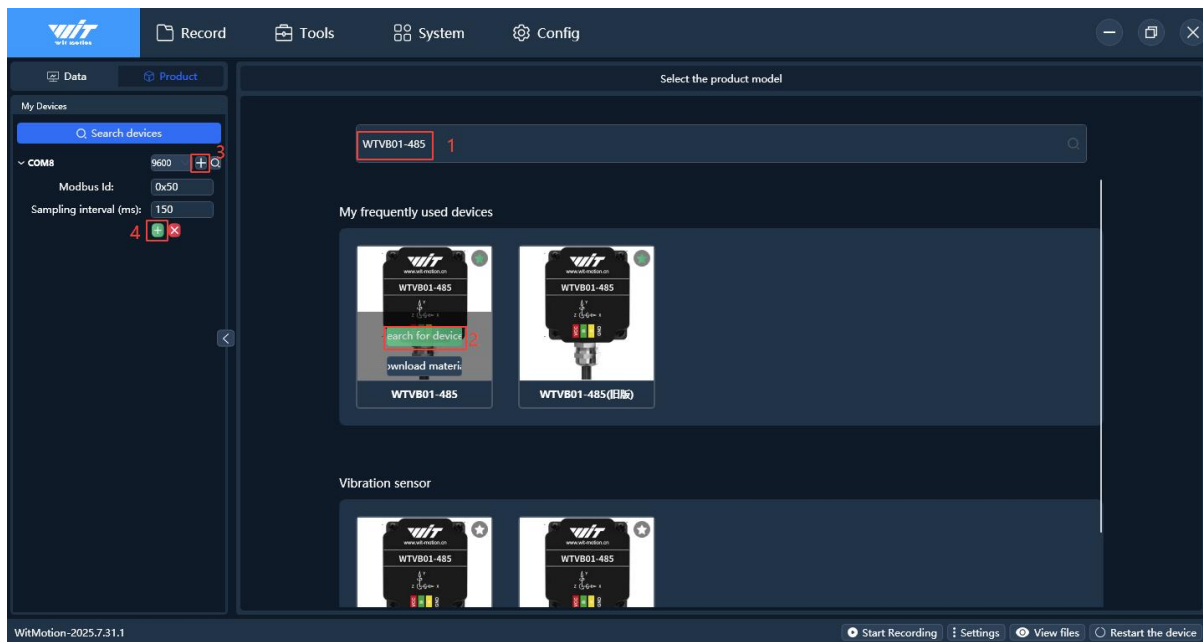
After the connection is successful, the data will be displayed on the software, as shown below:



7.1.2. Manual Connection

1. Select the model number WTVB01-485.
2. Select the baud rate of the corresponding port. The default baud rate is 9600
3. Enter the corresponding ID (the default is 0x50) and click the + sign to add the device.

As shown in the figure below:



7.2. Graph

Click on the graph to see graphs of vibration speed, vibration angle, vibration displacement and temperature (normal mode); in high-speed mode, only the vibration displacement graph has data, as shown below:



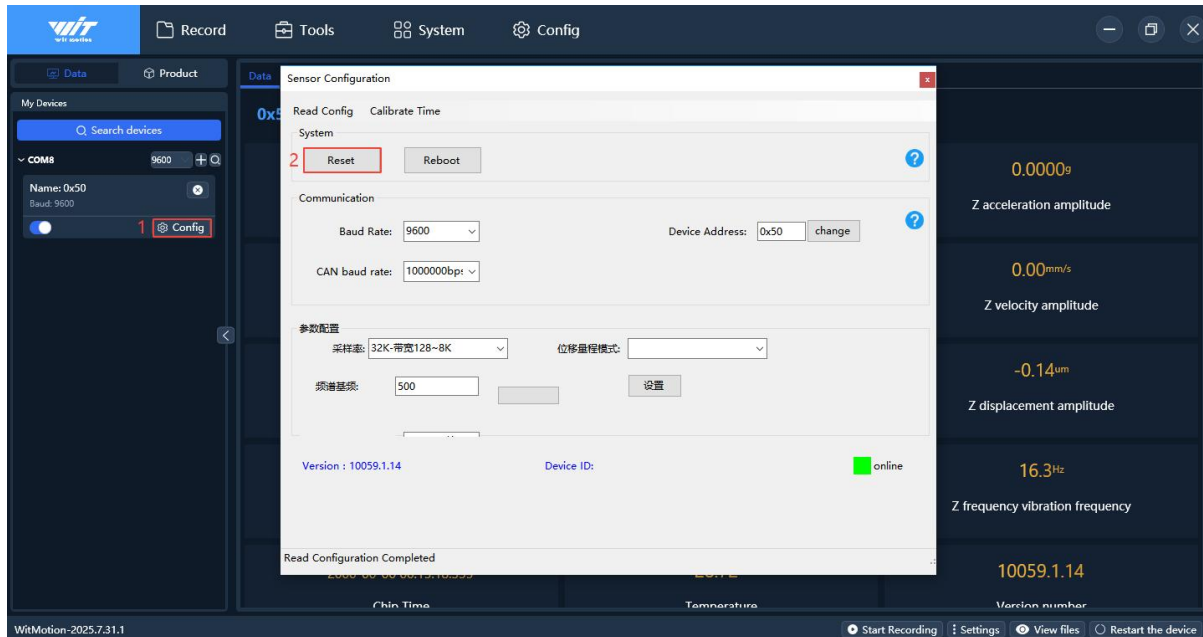
7.3. Scatter plots

If you need to obtain a complete vibration displacement scatter plot, it is recommended to use high-speed mode for viewing.



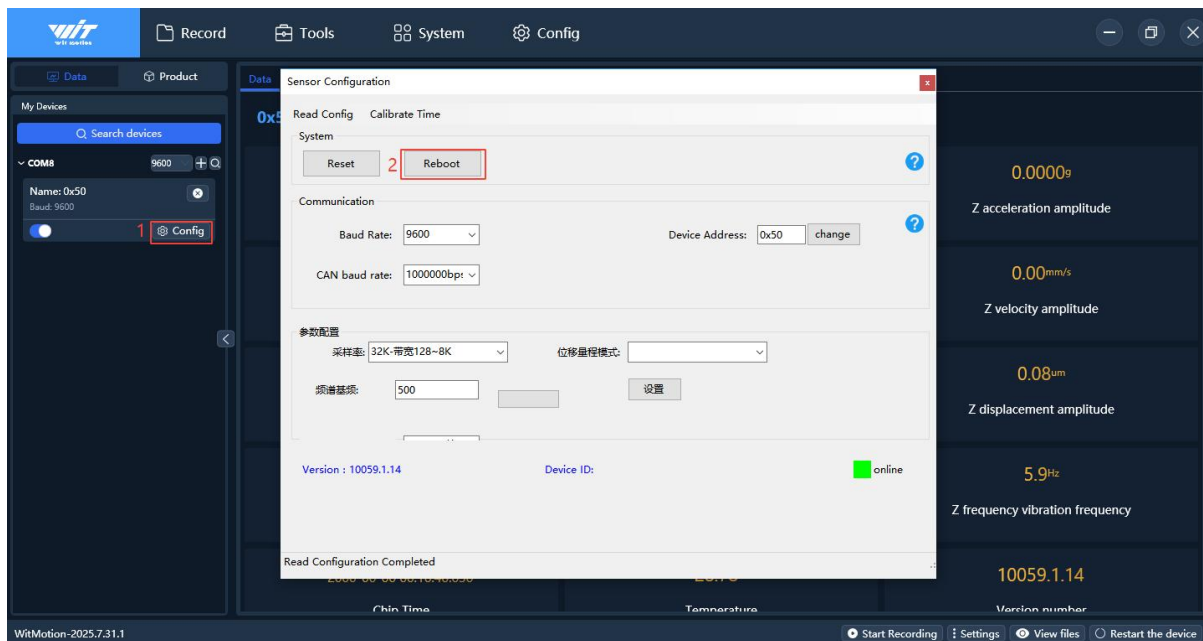
8. Software Configuration

9. Restore settings



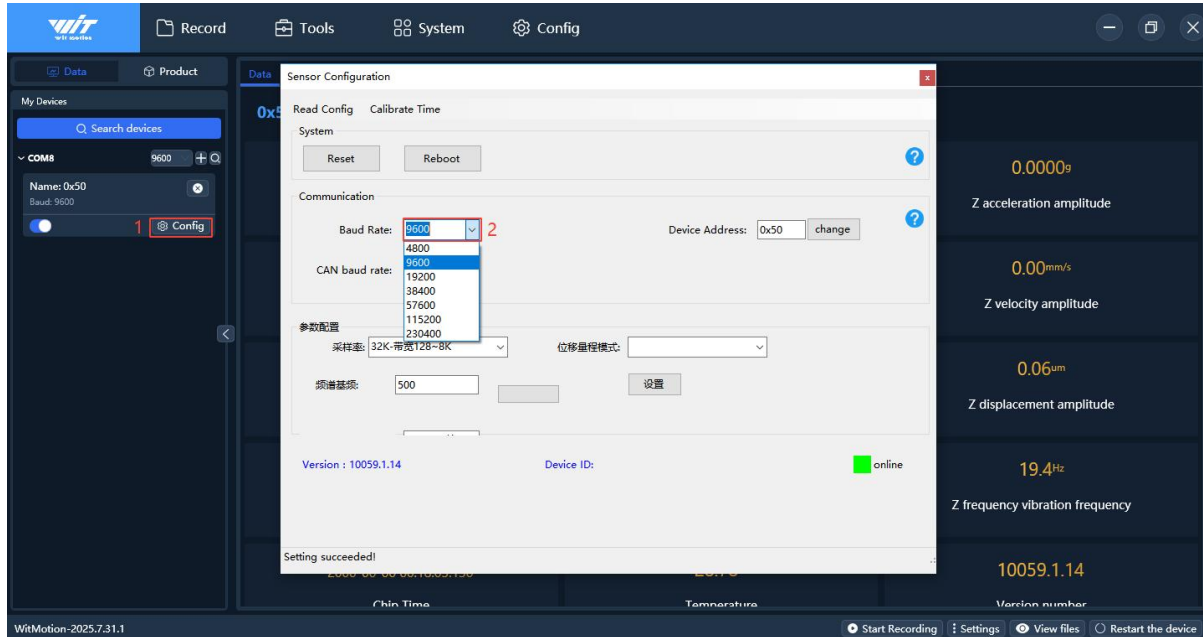
9.1. Restart

Click "Config" and then click "Reboot" in the sensor configuration interface to restart the sensor, as shown below:



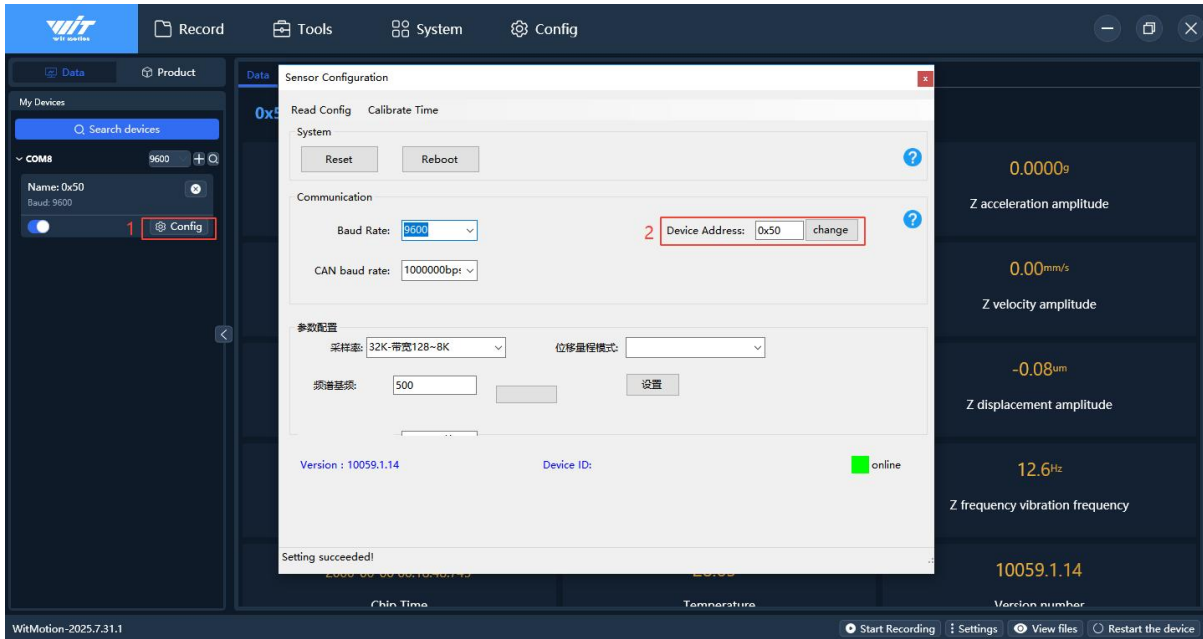
9.2. Communication rate

Click "Config" and click the "Baud Rate" drop-down menu in the sensor configuration interface. Select the serial port baud rate to be modified to change the current serial port baud rate (the default serial port baud rate is 9600). The serial port baud rate can be 4800, 9600, 19200, 38400, 57600, 115200, and 230400, as shown in the figure below:



9.3. Device Address

Click "Config" and click the "Device Address" input box in the sensor configuration interface. Enter the Device address and click Set to change the Device address (the default Device address is 0x50). The Device address ranges from 0x00 to 0x7F, as shown in the figure:



10. Communication Protocol

Protocol: MODBUS protocol

Level: 485 level (default baud rate: 9600bps)

10.1. Read register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address, register count, and data read are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, the high and low bits of the register count are represented by LEN_H and LEN_L, and the high and low bits of the data read are represented by DATA_{1H} and DATA_{1L}.
- The last two digits of the read instruction are standard CRC check digits.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	Read the high 8 bits of the length	Read the lower 8 bits of the length	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	ADDR _H [15:8]	ADDR _L [7:0]	LEN _H [3:08 PM]	LEN _L [7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Read length	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x03 (Read)	LEN[7:0]	DATA1H[15:8]	DATA1L[7:0]	...	...	DATANH	DATANL	CRCH[15:8]	CRCL[7:0]

10.2. Write register format

- The data is sent in hexadecimal format, not ASCII.
- Each register address and write data are represented by two bytes. The high and low bits of the register address are represented by ADDR_H and ADDR_L, and the high and low bits of the write data are represented by DATA_H and DATA_L.

Command sending

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Data return

Modbus Address	Function code	Register high 8 bits	Register lower 8 bits	High 8 bits of data	Data lower 8 bits	Check digit high 8 bits	Check digit lower 8 bits
ID	0x06 (Write)	ADDRH[15:8]	ADDRL[7:0]	DATAH[15:8]	DATAL[7:0]	CRCH[15:8]	CRCL[7:0]

Note:

The instruction writing operation needs to be performed in three steps.

The first step is to unlock 0x50 0x06 0x00 0x69 0xB5 0x88 0x22 0xA1. The unlocking will take effect within ten seconds.

The second step is to send the instructions that need to be modified.

The third step is to save the instruction 0x50 0x06 0x00 0x00 0x00 0x84 0x4B.

10.3. Register address table

Reg addr	Symbol	Meaning	Units and Explanations
0x00	SAVE	Save/Reboot/Restore Factory Settings	
0x04	BAUD	Serial port baud rate	
0x1A	IICADDR	Device address	
0x30	YYMM	Monthly Year	
0x31	DDHH	Time	
0x32	MMSS	Seconds and minutes	
0x33	MS	millisecond	
0x34	AX	X-axis acceleration	g
0x35	AY	Y-axis acceleration	g
0x36	AZ	Z- axis acceleration	g
0x3A	VX	X-axis vibration velocity	mm/s
0x3B	VY	Y-axis vibration velocity	mm/s
0x3C	VZ	Z-axis vibration velocity	mm/s
0x40	TEMP	Chip temperature	°C
0x41	DX	X-axis vibration displacement	mm
0x42	DY	Y-axis vibration displacement	mm

0x43	DZ	Z-axis vibration displacement	mm
0x44	HZX	X-axis vibration frequency	Hz
0x45	HZY	Y-axis vibration frequency	Hz
0x46	HZZ	Z-axis vibration frequency	Hz
0x47	CFX	X-axis acceleration peak factor	The peak value to RMS ratio measures the degree of signal abrupt change (g).
0x48	KX	X-axis acceleration kurtosis value	Detection of impact components or abnormal vibrations g
0x49	AAVGX	Absolute average acceleration along the X-axis	The average of the absolute values of vibrations measures the overall intensity g.
0x4A	VARX	X-axis acceleration variance	g reflects the magnitude of the signal fluctuation range
0x4B	RRAX	X-axis acceleration root square magnitude (RMS)	The standard index of energy intensity (g)
0x4C	Wix	X-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape.
0x50	PIX	Peak X-axis acceleration	The ratio of peak value to absolute average value g
0x51	VKX	X-axis velocity kurtosis value	Speed signal impact detection g
0x52	DRMSX	X-axis displacement RMS value	Measurement of displacement change intensity (mm)
0x53	CFY	Y-axis acceleration peak factor	The peak value to RMS ratio measures the degree of signal abrupt change (g).

0x54	KY	Y-axis acceleration kurtosis value	Detection of impact components or abnormal vibrations g
0x55	AAVG	Absolute average Y-axis acceleration	The average of the absolute values of vibrations measures the overall intensity g.
0x56	VARY	Y-axis acceleration variance	g reflects the magnitude of the signal fluctuation range
0x57	RRAY	Y-axis acceleration root square magnitude (RMS)	The standard index of energy intensity (g)
0x58	WIY	Y-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape.
0x59	PIY	Y-axis acceleration peak index	The ratio of peak value to absolute average value g
0x5A	PCY	Y-axis acceleration pulse coefficient	The ratio of peak value to average amplitude
0x5B	SKY	Y-axis acceleration skewness	Measure signal symmetry (positive or negative bias) g
0x5C	VRMSY	Y-axis velocity RMS value	Velocity signal energy intensity (mm/s)
0x5D	VKY	Y-axis velocity kurtosis value	Speed signal impact detection g
0x5E	DRMSY	Y-axis displacement RMS value	Measurement of displacement change intensity (mm)
0x5F	CFZ	Z-axis acceleration peak factor	The peak value to RMS ratio measures the degree of signal abrupt change (g).

0x60	KZ	Z-axis acceleration kurtosis value	Detection of impact components or abnormal vibrations g
0x61	AAVGZ	Absolute average Z-axis acceleration	The average of the absolute values of vibrations measures the overall intensity g.
0x62	VARZ	Z-axis acceleration variance	g reflects the magnitude of the signal fluctuation range
0x63	RRAZ	Z-axis acceleration root square magnitude (RMS)	The standard index of energy intensity (g)
0x64	WIZ	Z-axis acceleration waveform index	The ratio of RMS to the absolute mean reflects the waveform shape.
0x65	PIZ	Z-axis acceleration peak index	The ratio of peak value to absolute average value g
0x66	PCZ	Z-axis acceleration pulse coefficient	The ratio of peak value to average amplitude
0x67	SKZ	Z-axis acceleration skewness	The measure of signal symmetry (positive or negative bias) g
0x68	VRMSZ	Z-axis speed RMS value	Velocity signal energy intensity (mm/s)
0x69	VKZ	Z-axis velocity kurtosis value	Speed signal impact detection g
0x6A	DRMSZ	Z-axis displacement RMS value	Measurement of displacement change intensity (mm)
0x6B	ERRX	X-axis fault diagnosis	
0x6C	ERRY	Y-axis fault diagnosis	
0x6D	ERRZ	Z-axis fault diagnosis	
0x6E	AX_ENERG Y_POINT1	X-axis focus 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics

0x6F	AX_ENERG Y_POINT2	X-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals using half harmonics
0x70	AX_ENERG Y_POINT3	X-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x71	AX_ENERG Y_POINT4	X-axis focus point 4: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (second harmonics)
0x72	AX_ENERG Y_POINT5	X-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x73	AX_ENERG Y_POINT6	X-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)
0x74	AX_ENERG Y_POINT7	X-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)
0x75	AX_ENERG Y_POINT8	X-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x76	AX_ENERG Y_AREA1	X-axis focus band 1 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 0 to 1/2 harmonic.
0x77	AX_ENERG Y_AREA2	X-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.
0x78	AX_ENERG Y_AREA3	X-axis focus on the acceleration spectrum energy of band 3	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x79	AX_ENERG Y_AREA4	X-axis focus on band 4 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x7A	AX_ENERG Y_AREA5	X-axis focus on the acceleration spectrum energy of band 5	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics

0x7B	AX_ENERG Y_AREA6	X-axis focus on the acceleration spectrum energy of band 6	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x7C	AX_ENERG Y	Total energy of X-acceleration spectrum	The sum of acceleration energy across all frequency ranges
0x7D	AY_ENERG Y_POINT1	Y-axis focus point 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics
0x7E	AY_ENERG Y_POINT2	Y-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at half harmonics
0x7F	AY_ENERG Y_POINT3	Y-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x80	AY_ENERG Y_POINT4	Y-axis focus point 4: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (second harmonics)
0x81	AY_ENERG Y_POINT5	Y-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x82	AY_ENERG Y_POINT6	Y-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)
0x83	AY_ENERG Y_POINT7	Y-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)
0x84	AY_ENERG Y_POINT8	Y-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x85	AY_ENERG Y_AREA1	Y-axis focus band 1 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 0 to 1/2 harmonic.
0x86	AY_ENERG Y_AREA2	Y-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.

0x87	AY_ENERG Y_AREA3	Y-axis focus band 3 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x88	AY_ENERG Y_AREA4	Y-axis focus band 4 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x89	AY_ENERG Y_AREA5	Y-axis focus on the acceleration spectrum energy of band 5	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics
0x8A	AY_ENERG Y_AREA6	Y-axis focus on the acceleration spectrum energy of band 6	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x8B	AY_ENERG Y	Total energy of Y acceleration spectrum	The sum of acceleration energy across all frequency ranges
0x8C	AZ_ENERG Y_POINT1	Z-axis focus point 1: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at quarter harmonics
0x8D	AZ_ENERG Y_POINT2	Z-axis focus point 2: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals at half harmonics
0x8E	AZ_ENERG Y_POINT3	Z-axis focus point 3: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (first harmonic)
0x8F	AZ_ENERG Y_POINT4	Z-axis focus point 4: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (second harmonics)
0x90	AZ_ENERG Y_POINT5	Z-axis focus point 5: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (third harmonic)
0x91	AZ_ENERG Y_POINT6	Z-axis focus point 6: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fourth harmonic)
0x92	AZ_ENERG Y_POINT7	Z-axis focus point 7: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (fifth harmonic)

0x93	AZ_ENERGY_POINT8	Z-axis focus point 8: Acceleration spectrum energy	Energy analysis of key frequency points of acceleration signals (sixth harmonic)
0x94	AZ_ENERGY_AREA1	Z-axis focus band 1 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 0 to 1/2 harmonic.
0x95	AZ_ENERGY_AREA2	Z-axis focus band 2 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band from 1/2 to 1 harmonic.
0x96	AZ_ENERGY_AREA3	Z-axis focus zone 3 acceleration spectrum energy	Energy monitoring of acceleration signals in the frequency band (1st to 2nd harmonic)
0x97	AZ_ENERGY_AREA4	Z-axis focus on the acceleration spectrum energy of band 4	Energy monitoring of acceleration signals in the frequency band at 2-3 harmonics
0x98	AZ_ENERGY_AREA5	Z-axis focus on the acceleration spectrum energy of band 5	Energy monitoring of acceleration signals in the frequency band at 3-4 harmonics
0x99	AZ_ENERGY_AREA6	Z-axis focus on acceleration spectrum energy (band 6)	Energy monitoring of acceleration signals in the frequency band at 4-5 harmonics
0x9A	AZ_ENERGY	Total energy of the Z-acceleration spectrum	The sum of acceleration energy across all frequency ranges

10.4. Register Description

All the examples below are instructions with Modbus address 0x50 (default). If you change the Modbus address, you need to change the address and CRC check bit in the instructions accordingly.

10.4.1. SAVE (Save/Reboot/Restore Factory Settings)

Register name: SAVE

Register address: 0 (0x00)

Read/W direction: R/W

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	SAVE[15:0]	Saved: 0x0000 Reboot: 0x00FF Factory reset: 0x0001

Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful for unlocking within 10 seconds)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 00 00 FF C4 0B (Reboot)

Return: 50 06 00 00 00 FF C4 0B

10.4.2. BAUD (Serial Baud Rate)

Register name: BAUD

Register address: 4 (0x04)

Read/W direction: R/W

Default value: 0x0002

Bit	NAME	FUNCTION
15:4		

3:0	BAUD[3:0]	Set the serial port baud rate: 0001(0x01) 4800bps 0010(0x02): 9600bps 0011(0x03): 19200bps 0100(0x04): 38400bps 0101(0x05): 57600bps 0110(0x06): 115200bps 0111(0x07): 230400bps
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Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful for unlocking within 10 seconds)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 04 00 06 45 88 (Set serial port baud rate to 115200)

Send: 50 06 00 00 00 00 84 4B (Save at 115200 baud)

Return: 50 06 00 00 00 00 84 4B

10.4.3. IICADDR (Device Address)

Register name: IICADDR

Register address: 26 (0x1A)

Read/W direction: R/W

Default value: 0x0050

Bit	NAME	FUNCTION
15:8		
7:0	IICADDR[7:0]	Set the device address, used for I2C and Modbus

communication, using 0x01~0x7F.

Example:

Send: 50 06 00 69 B5 88 22 A1 (Useful for unlocking within 10 seconds)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 1A **00 02** 24 4D (Set device address to **0x02**)

Return: 50 06 00 1A 00 02 24 4D

Send: **02** 06 00 69 B5 88 2F 13 (Useful for unlocking within 10 seconds)

Return: 02 06 00 69 B5 88 2F 13

Send: **02** 06 00 00 00 00 89 F9 (Save)

Return: 02 06 00 00 00 00 89 F9

10.4.4. YYMM~MS (on-screen time)

Register names: YYMM~MS

Register addresses: 48~51 (0x30~0x33)

Read/W direction: R/W

Default time: (2015, 1, 1, 00, 00, 59, 00)

Bit	NAME	FUNCTION
15:8	YYMM[15:8]	moon
7:0	YYMM[7:0]	Year
15:8	DDHH[15:8]	hour

7:0	DDHH[7:0]	day
15:8	MMSS[15:8]	Second
7:0	MMSS[7:0]	point
15:0	MS[15:0]	millisecond

Example: Read

Send: 50 03 00 30 00

Example: Write settings

Send: 50 06 00 69 B5 88 22 A1 (Useful for unlocking within 10 seconds)

Return: 50 06 00 69 B5 88 22 A1

Send: 50 06 00 30 03 16 05 7A (Set the year and month as 22-03)

Return: 50 06 00 30 03 16 05 7A

Send: 50 06 00 31 09 0C D3 D1 (Set date and time 12-09)

Return: 50 06 00 31 09 0C D3 D1

Send: 50 06 00 32 3A 1E B7 2C (Set minutes and seconds 30:58)

Return: 50 06 00 32 3A 1E B7 2C

Send: 50 06 00 33 01 F4 74 53 (Set milliseconds to 500)

Return: 50 06 00 33 01 F4 74 53

Send: 50 06 00 00 00 00 84 4B (Save)

Return: 50 06 00 00 00 00 84 4B

10.4.5. AX~AZ (acceleration)

Register names: AX~AZ

Register addresses: 52~54 (0x34~0x36)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX[15:0]	Acceleration X = $AX[15:0]/32768 \times 16g$ (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)
15:0	AY[15:0]	Acceleration Y = $AY[15:0]/32768 \times 16g$ (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)
15:0	AZ[15:0]	Acceleration Z = $AZ[15:0]/32768 \times 16g$ (g is the acceleration due to gravity, which can be taken as 9.8m/s ²)

Example:

Send: 50 03 00 34 00 03 49 84 (Read triaxial acceleration)

Return: 50 03 06 AXH AXL AYH AYL AZH AZL CRCH CRCL

$AX[15:0] = ((\text{short})AXH \ll 8) | AXL;$

$AY[15:0] = ((\text{short})AYH \ll 8) | AYL;$

$AZ[15:0] = ((\text{short})AZH \ll 8) | AZL;$

10.4.6. VX~VZ (Vibration Velocity)

Register names: VX~VZ

Register addresses: 58~60 (0x3A~0x3C)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	VX[15:0]	Vibration velocity VX (mm/s) = ((VXH << 8) VXL)/100
15:0	VY[15:0]	Vibration velocity VY (mm/s) = ((VYH << 8) VYL)/100
15:0	VZ[15:0]	Vibration velocity VZ (mm/s) = ((VZH << 8) VZL)/100

Example:

Send: 50 03 00 3A 00 03 28 47 (Read the triaxial vibration speed)

Returns: 50 03 06 VXH VXL VYH VYL VZH VZL CRCH CRCL

$VX[15:0] = (((\text{short})VXH \ll 8) | VXL) / 100;$

$VY[15:0] = (((\text{short})VYH \ll 8) | VYL) / 100;$

$VZ[15:0] = (((\text{short})VZH \ll 8) | VZL) / 100;$

10.4.7. Reserved

Register Name: Reserved

Register addresses: 61~63 (0x3D~0x3F)

Reading/Writing Direction: R

Default value: 0x0000

10.4.8. TEMP (temperature)

Register name: TEMP

Register address: 64 (0x40)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	TEMP[15:0]	Temperature = TEMP[15:0] / 100°C

Example:

Send: 50 03 00 40 00 01 88 5F (Read chip temperature)

Return: 50 03 02 TEMPH TEMPL CRCH CRCL

TEMP[15:0]=(((short)TEMPH <<8)|TEMPL);

10.4.9. DX~DZ (Vibration Displacement)

Register names: DX~DZ

Register addresses: 65~67 (0x41~0x43)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
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15:0	DX[15:0]	Vibration displacement DX(um) = ((DXH << 8) DXL)
15:0	DY[15:0]	Vibration displacement DY(um) = ((DYH << 8) DYL)
15:0	DZ[15:0]	Vibration displacement DZ(um) = ((DZH << 8) DZL)

Example:

Send: 50 03 00 41 00 03 58 5E (Read triaxial vibration displacement)

Return: 50 03 06 DXH DXL DYH DYL DZH DZL CRCH CRCL

DX[15:0]=(((short)DXH <<8)|DXL);

DY[15:0]=(((short)DYH <<8)|DYL);

DZ[15:0]=(((short)DZH <<8)|DZL);

10.4.10. HZX ~ HZZ (Vibration frequency)

Register names: HZX~HZZ

Register addresses: 68~70 (0x44~0x46)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	HZX[15:0]	Vibration frequency HZX (Hz) = ((HZXH << 8) HZXL)/10
15:0	HZY[15:0]	Vibration frequency HZY (Hz) = ((HZYH << 8) HZYL)/10
15:0	HZZ[15:0]	Vibration frequency HZZ (Hz) = ((HZZH << 8) HZZL)/10

Example:

Send: 50 03 00 44 00 03 48 5F (Read the triaxial vibration frequency)

Returns: 50 03 06 HZXH HZXL HZYH HZYL HZZH HZZL CRCH CRCL

$HZX[15:0] = (((\text{short})HZXH \ll 8) | HZXL) / 10;$

$HZY[15:0] = (((\text{short})HZYH \ll 8) | HZYL) / 10;$

$HZZ[15:0] = (((\text{short})HZZH \ll 8) | HZZL) / 10;$

10.4.11. X-axis vibration simulation quantity

Register names: CFX~DRMSX

Register addresses: 71~82 (0x47~0x52)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFX[15:0]	X-axis acceleration peak factor CFX(g) = ((CFXH << 8) CFXL)/1000
15:0	KX [15:0]	X-axis acceleration kurtosis value KX (g) = ((KXH << 8) KXL) / 1000
15:0	AAVGX [15:0]	The absolute average value of the X-axis acceleration, AAVGX (g), is calculated as follows: AAVGXH << 8 AAVGXL)/1000
15:0	VARX [15:0]	X-axis acceleration variance VARX (g) = ((VARXH << 8 VARXL)/1000
15:0	RRAX [15:0]	X-axis acceleration root square magnitude (RMS) RRAX (g) = ((RRAXH << 8 RRAXL)/1000

15:0	WIX [15:0]	X-axis acceleration waveform index $WIX = ((WIXH \ll 8 WIXL) / 1000$
15:0	PIX [15:0]	The peak X-axis acceleration index $PIX (g) = ((PIXH \ll 8 PIXL) / 1000$
15:0	PCX [15:0]	X-axis acceleration pulse coefficient $PCX = ((PCXH \ll 8 PCXL) / 1000$
15:0	SKX [15:0]	X-axis acceleration skewness $SKX (g) = ((SKXH \ll 8 SKXL) / 1000$
15:0	VRMSX [15:0]	X-axis velocity RMS value $VRMSX (mm/s) = ((VRMSXH \ll 8 VRMSXL) / 1000$
15:0	VKX [15:0]	X-axis velocity kurtosis value $VKX (g) = ((VKXH \ll 8 VKXL) / 1000$
15:0	DRMSX [15:0]	X-axis displacement RMS value $DRMSX (mm) = ((DRMSXH \ll 8 DRMSXL) / 1000$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

$NAME[15:0] = (((short)NAMEH \ll 8 | NAMEL) / 1000;$

Example:

Send: 50 03 00 47 00 01 39 9E (X-axis acceleration peak factor)

Return: 50 03 06 CFXH CFXL CRCH CRCL

$CFX[15:0] = (((short)CFXH \ll 8 | CFXL) / 1000;$

10.4.12. Y-axis vibration simulation quantity

Register names: CFY~DRMSY

Register addresses: 83~94 (0x53~0x5E)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFY[15:0]	Y-axis acceleration peak factor CFY(g) = ((CFYH << 8) CFYL)/1000
15:0	KY [15:0]	Y-axis acceleration kurtosis KY (g) = ((KYH << 8) KYL) / 1000
15:0	AAVGY [15:0]	The absolute average value of the Y-axis acceleration, AAVGY (g), is calculated as follows: AAVGYH << 8 AAVGYL)/1000
15:0	VARY [15:0]	Y-axis acceleration variance VARY (g) = ((VARYH << 8 VARYL)/1000
15:0	RRAY [15:0]	Y-axis acceleration root square magnitude (RMS) RRAY (g) = ((RRAYH << 8 RRAYL)/1000
15:0	WIY [15:0]	Y-axis acceleration waveform index WIY = ((WIYH << 8 WIYL)/1000
15:0	PIY [15:0]	The peak Y-axis acceleration index PIY (g) = ((PIYH << 8 PIYL)/1000
15:0	PCY [15:0]	Y-axis acceleration pulse coefficient PCY = ((PCYH << 8 PCYL)/1000
15:0	SKY [15:0]	Y-axis acceleration skewness SKY (g) = ((SKYH << 8 SKYL)/1000
15:0	VRMSY [15:0]	Y-axis velocity RMS value VRMSY (mm/s) = ((VRMSYH << 8 VRMSYL)/1000
15:0	VKY [15:0]	Y-axis velocity kurtosis value VKY (g) = ((VKYH << 8 VKYL)/1000
15:0	DRMSY [15:0]	Y-axis displacement RMS value DRMSY (mm) = ((DRMSYH << 8 DRMSYL)/1000

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 53 00 01 79 9A (Y-axis acceleration peak factor)

Return: 50 03 06 CFYH CFYL CRCH CRCL

CFX[15:0]=(((short)CFYH <<8)|CFYL)/1000;

10.4.13. Z-axis vibration simulation quantity

Register names: CFZ~DRMSZ

Register addresses: 95~106 (0x5F~0x6A)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	CFZ[15:0]	Z-axis acceleration peak factor CFZ(g) = ((CFZH << 8) CFZL)/1000
15:0	KZ [15:0]	Z-axis acceleration kurtosis value KZ (g) = ((KZH << 8) KZL)/1000
15:0	AAVGZ[15:0]	The absolute average value of the Z-axis acceleration, AAVGZ (g), is calculated as follows: AAVGZH << 8 AAVGZL)/1000

15:0	VARZ [15:0]	The variance of Z-axis acceleration, VARZ (g), is calculated as follows: $VARZ = ((VARZH \ll 8 VARZL) / 1000)$
15:0	RRAZ [15:0]	Z-axis acceleration root mean square magnitude (RMS) RRAZ (g) = $((RRAZH \ll 8 RRAZL) / 1000)$
15:0	WIZ [15:0]	Z-axis acceleration waveform index WIZ = $((WIZH \ll 8 WIZL) / 1000)$
15:0	PIZ [15:0]	Z-axis peak acceleration index PIZ (g) = $((PIZH \ll 8 PIZL) / 1000)$
15:0	PCZ [15:0]	Z-axis acceleration pulse coefficient PCZ = $((PCZH \ll 8 PCZL) / 1000)$
15:0	SKZ [15:0]	Z-axis acceleration skewness SKZ (g) = $((SKZH \ll 8 SKZL) / 1000)$
15:0	VRMSZ [15:0]	Z-axis velocity RMS value VRMSZ (mm/s) = $((VRMSZH \ll 8 VRMSZL) / 1000)$
15:0	VKZ [15:0]	Z-axis velocity kurtosis value VKZg) = $((VKZH \ll 8 VKZL) / 1000)$
15:0	DRMSZ [15:0]	Z-axis displacement RMS value DRMSZ (mm) = $((DRMSZH \ll 8 DRMSZL) / 1000)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL)/1000;

Example:

Send: 50 03 00 5F 00 01 79 9A (Y-axis acceleration peak factor)

Return: 50 03 06 CFZH CFZL CRCH CRCL

CFZ[15:0]=(((short)CFZH <<8)|CFZL)/1000;

10.4.14. XYZ axis fault diagnosis

Register names: ERRX~ERRZ

Register addresses: 107~109 (0x6B~0x6D)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	ERRX	
15:0	ERRY	
15:0	ERRZ	

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 6B 00 01 F8 57

Return: 50 03 06 ERRXH ERRXL CRCH CRCL

ERRX[15:0]=(((short)ERRXH <<8)|ERRXL);

10.4.15. X-axis focus and acc spectrum energy

Register names: AX_ENERGY_POINT1 ~ AX_ENERGY

Register addresses: 110~124 (0x6E~0x7C)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AX_ENERGY_POINT1[15:0]	X-axis focus point 1: Acceleration spectrum energy AX_ENERGY_POINT1 = ((AX_ENERGY_POINT1H << 8) AX_ENERGY_POINT1L)
15:0	AX_ENERGY_POINT2 [15:0]	X-axis focus point 2: Acceleration spectrum energy AX_ENERGY_POINT2 = ((AX_ENERGY_POINT2H << 8) AX_ENERGY_POINT2L)
15:0	AX_ENERGY_POINT3[15:0]	X-axis focus point 3: Acceleration spectrum energy AX_ENERGY_POINT3 = ((AX_ENERGY_POINT3H << 8) AX_ENERGY_POINT3L)
15:0	AX_ENERGY_POINT4 [15:0]	X-axis focus point 4: Acceleration spectrum energy AX_ENERGY_POINT1 = ((AX_ENERGY_POINT4H << 8) AX_ENERGY_POINT4L)
15:0	AX_ENERGY_POINT5 [15:0]	X-axis focus point 5: Acceleration spectrum energy AX_ENERGY_POINT5 = ((AX_ENERGY_POINT5H << 8) AX_ENERGY_POINT5L)
15:0	AX_ENERGY_POINT6 [15:0]	X-axis focus point 6: Acceleration spectrum energy AX_ENERGY_POINT6 =

		$((AX_ENERGY_POINT6H \ll 8) \mid AX_ENERGY_POINT6L)$
15:0	AX_ENERGY_POINT7 [15:0]	X-axis focus point 7: Acceleration spectrum energy AX_ENERGY_POINT7 = $((AX_ENERGY_POINT7H \ll 8) \mid AX_ENERGY_POINT7L)$
15:0	AX_ENERGY_POINT8 [15:0]	X-axis focus point 8: Acceleration spectrum energy AX_ENERGY_POINT8 = $((AX_ENERGY_POINT8H \ll 8) \mid AX_ENERGY_POINT8L)$
15:0	AX_ENERGY_AREA1 [15:0]	X-axis focus band 1 acceleration spectrum energy AX_ENERGY_AREA1 = $((AX_ENERGY_AREA1H \ll 8) \mid AX_ENERGY_AREA1L)$
15:0	AX_ENERGY_AREA2 [15:0]	X-axis focus band 2 acceleration spectrum energy AX_ENERGY_AREA2 = $((AX_ENERGY_AREA2H \ll 8) \mid AX_ENERGY_AREA2L)$
15:0	AX_ENERGY_AREA3 [15:0]	X-axis focus band 3 acceleration spectrum energy AX_ENERGY_AREA3 = $((AX_ENERGY_AREA3H \ll 8) \mid AX_ENERGY_AREA3L)$
15:0	AX_ENERGY_AREA4 [15:0]	X-axis focus band 4 acceleration spectrum energy AX_ENERGY_AREA4 = $((AX_ENERGY_AREA4H \ll 8) \mid AX_ENERGY_AREA4L)$
15:0	AX_ENERGY_AREA5 [15:0]	X-axis focus band 5 acceleration spectrum energy AX_ENERGY_AREA5 = $((AX_ENERGY_AREA5H \ll 8) \mid AX_ENERGY_AREA5L)$
15:0	AX_ENERGY_AREA6 [15:0]	X-axis focus band 6 acceleration spectrum energy AX_ENERGY_AREA6 = $((AX_ENERGY_AREA6H \ll 8) \mid AX_ENERGY_AREA6L)$
15:0	AX_ENERGY [15:0]	The total energy of the X-axis acceleration spectrum is AX_ENERGY = $((AX_ENERGYH \ll 8) \mid AX_ENERGYL)$

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

```

Send: 50 03 00 6E 00 01 E8 56 ( X-axis focus point 1: Acceleration spectrum energy )

Return: 50 03 06 AX_ENERGY_POINT1H AX_ENERGY_POINT1L CRCH CRCL

AX_ENERGY_POINT1[15:0]=(((short)AX_ENERGY_POINT1H <<8)|AX_ENERGY_POINT1L);

```

10.4.16. Y-axis focus and acc spectrum energy

Register names: AY_ENERGY_POINT1~ AY_ENERGY

Register addresses: 125~139 (0x7D~0x8B)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AY_ENERGY_POINT1[15:0]	Y-axis focus point 1: Acceleration spectrum energy AY_ENERGY_POINT1 = ((AY_ENERGY_POINT1H << 8) AY_ENERGY_POINT1L)
15:0	AY_ENERGY_POINT2 [15:0]	Y-axis focus point 2: Acceleration spectrum energy AY_ENERGY_POINT2=((AY_ENERGY_POINT2H << 8) AY_ENERGY_POINT2L)
15:0	AY_ENERGY_POINT3[15:0]	Y-axis focus point 3: Acceleration spectrum energy AY_ENERGY_POINT3=((AY_ENERGY_POINT3H << 8) AY_ENERGY_POINT3L)
15:0	AY_ENERGY_POINT4 [15:0]	Y-axis focus point 4: Acceleration spectrum energy AY_ENERGY_POINT1=((AY_ENERGY_POINT4H << 8) AY_ENERGY_POINT4L)
15:0	AY_ENERGY_POINT5 [15:0]	Y-axis focus point 5: Acceleration spectrum energy AY_ENERGY_POINT5 = ((AY_ENERGY_POINT5H << 8) AY_ENERGY_POINT5L)

15:0	AY_ENERGY_POINT6 [15:0]	Y-axis focus point 6: Acceleration spectrum energy AY_ENERGY_POINT6 = ((AY_ENERGY_POINT6H << 8) AY_ENERGY_POINT6L)
15:0	AY_ENERGY_POINT7 [15:0]	Y-axis focus point 7: Acceleration spectrum energy AY_ENERGY_POINT7 = ((AY_ENERGY_POINT7H << 8) AY_ENERGY_POINT7L)
15:0	AY_ENERGY_POINT8 [15:0]	Y-axis focus point 8: Acceleration spectrum energy AY_ENERGY_POINT8 = ((AY_ENERGY_POINT8H << 8) AY_ENERGY_POINT8L)
15:0	AY_ENERGY_AREA1 [15:0]	Y-axis focus band 1 acceleration spectrum energy AY_ENERGY_AREA1 = ((AY_ENERGY_AREA1H << 8 AY_ENERGY_AREA1L)
15:0	AY_ENERGY_AREA2 [15:0]	Y-axis focus band 2 acceleration spectrum energy AY_ENERGY_AREA2 = ((AY_ENERGY_AREA2H << 8 AY_ENERGY_AREA2L)
15:0	AY_ENERGY_AREA3 [15:0]	Y-axis focus band 3 acceleration spectrum energy AY_ENERGY_AREA3 = ((AY_ENERGY_AREA3H << 8 AY_ENERGY_AREA3L)
15:0	AY_ENERGY_AREA4 [15:0]	Y-axis focus band 4 acceleration spectrum energy AY_ENERGY_AREA4 = ((AY_ENERGY_AREA4H << 8 AY_ENERGY_AREA4L)
15:0	AY_ENERGY_AREA5 [15:0]	Y-axis focus band 5 acceleration spectrum energy AY_ENERGY_AREA5 = ((AY_ENERGY_AREA5H << 8 AY_ENERGY_AREA5L)
15:0	AY_ENERGY_AREA6 [15:0]	Y-axis focus band 6 acceleration spectrum energy AY_ENERGY_AREA6 = ((AY_ENERGY_AREA6H << 8 AY_ENERGY_AREA6L)
15:0	AY_ENERGY [15:0]	The total energy of the Y-axis acceleration spectrum is AY_ENERGY = ((AY_ENERGYH << 8 AY_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 7D 00 01 19 93 (Y-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AY_ENERGY_POINT1H AY_ENERGY_POINT1L CRCH CRCL

AY_ENERGY_POINT1[15:0]=(((short)AY_ENERGY_POINT1H <<8)|AY_ENERGY_POINT1L);

10.4.17. Z-axis focus and acc spectrum energy

Register names: AZ_ENERGY_POINT1 ~ AZ_ENERGY

Register addresses: 140~154 (0x8C~0x9A)

Reading/Writing Direction: R

Default value: 0x0000

Bit	NAME	FUNCTION
15:0	AZ_ENERGY_POINT1[15:0]	Z-axis focus point 1: Acceleration spectrum energy AZ_ENERGY_POINT1 = ((AZ_ENERGY_POINT1H << 8) AZ_ENERGY_POINT1L)
15:0	AZ_ENERGY_POINT2 [15:0]	Z-axis focus point 2: Acceleration spectrum energy AZ_ENERGY_POINT2 = ((AZ_ENERGY_POINT2H << 8) AZ_ENERGY_POINT2L)
15:0	AZ_ENERGY_POINT3[15:0]	Z-axis focus point 3: Acceleration spectrum energy AZ_ENERGY_POINT3 = ((AZ_ENERGY_POINT3H << 8) AZ_ENERGY_POINT3L)
15:0	AZ_ENERGY_POINT4 [15:0]	Z-axis focus point 4: Acceleration spectrum energy AZ_ENERGY_POINT1 = ((AZ_ENERGY_POINT4H << 8) AZ_ENERGY_POINT4L)

15:0	AZ_ENERGY_POINT5 [15:0]	Z-axis focus point 5: Acceleration spectrum energy AZ_ENERGY_POINT5 = ((AZ_ENERGY_POINT5H << 8) AZ_ENERGY_POINT5L)
15:0	AZ_ENERGY_POINT6 [15:0]	Z-axis focus point 6: Acceleration spectrum energy AY_ENERGY_POINT6 = ((AY_ENERGY_POINT6H << 8) AY_ENERGY_POINT6L)
15:0	AZ_ENERGY_POINT7 [15:0]	Z-axis focus point 7: Acceleration spectrum energy AY_ENERGZ_POINT7=((AZ_ENERGY_POINT7H << 8) AZ_ENERGY_POINT7L)
15:0	AZ_ENERGY_POINT8 [15:0]	Z-axis focus point 8: Acceleration spectrum energy AZ_ENERGY_POINT8 = ((AZ_ENERGY_POINT8H << 8) AZ_ENERGY_POINT8L)
15:0	AZ_ENERGY_AREA1 [15:0]	Z-axis focus band 1 acceleration spectrum energy AZ_ENERGY_AREA1 = ((AZ_ENERGY_AREA1H << 8 AZ_ENERGY_AREA1L)
15:0	AZ_ENERGY_AREA2 [15:0]	Y-axis focus band 2 acceleration spectrum energy AZ_ENERGY_AREA2 = ((AZ_ENERGY_AREA2H << 8 AZ_ENERGY_AREA2L)
15:0	AZ_ENERGY_AREA3 [15:0]	Z-axis focus zone 3 acceleration spectrum energy AZ_ENERGY_AREA3 = ((AZ_ENERGY_AREA3H << 8 AZ_ENERGY_AREA3L)
15:0	AZ_ENERGY_AREA4 [15:0]	Z-axis focus band 4 acceleration spectrum energy AZ_ENERGY_AREA4 = ((AZ_ENERGY_AREA4H << 8 AZ_ENERGY_AREA4L)
15:0	AZ_ENERGY_AREA5 [15:0]	Z-axis focus zone 5 acceleration spectrum energy AZ_ENERGY_AREA5 = ((AZ_ENERGY_AREA5H << 8 AZ_ENERGY_AREA5L)
15:0	AZ_ENERGY_AREA6 [15:0]	Z-axis focus zone 6 acceleration spectrum energy AZ_ENERGY_AREA6 = ((AZ_ENERGY_AREA6H << 8 AZ_ENERGY_AREA6L)
15:0	AZ_ENERGY [15:0]	The total energy of the Z-axis acceleration spectrum is AZ_ENERGY = ((AZ_ENERGYH << 8 AZ_ENERGYL)

Example:

Send: 50 03 00 NAME 00 01 CRCH CRCL

Return: 50 03 06 NAMEH NAMEL CRCH CRCL

NAME[15:0]=(((short)NAMEH <<8)|NAMEL);

Example:

Send: 50 03 00 8C 00 01 48 60 (Z-axis focus point 1: Acceleration spectrum energy)

Return: 50 03 06 AZ_ENERGY_POINT1H AZ_ENERGY_POINT1L CRCH CRCL

AZ_ENERGY_POINT1[15:0]=(((short)AZ_ENERGY_POINT1H <<8)|AZ_ENERGY_POINT1L);